

OCR Further Mathematics
Statistics - Correlation
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
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Question		Answer/Indicative content	Marks	Part marks and guidance	
1	a	The value of Pearson's pmcc would be changed by (most) such changes. The value of Spearman's r_s would not be changed as the ranks remain unchanged.	B1 (AO2.5) B1 (AO2.5) [2]	Explain effect on Pearson, or not known bivariate normal or not testing for linear correlation Explain why no effect on Spearman (not "not likely to be affected", or "not much affected" or "association not correlation")	
	b	H_0: no association between ranks of numbers of items H_1: (positive) association between ranks Ranks 1 2 3 4 5 6 7 8 9 4 1 3 2 8 5 9 7 6 $\Sigma d^2 = 38$ $r_s = 1 - \frac{6\Sigma d^2}{9(9^2 - 1)}$ = 0.683 < 0.700 Do not reject H_0. Insufficient evidence of association between rankings of the items	B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) M1 (AO1.2) A1 (AO1.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [8]	Don't insist on "population" here, but allow use of p_s in both, even if no explanation (not just r_s). Context needed, but don't worry about 1- or 2-tailed here Compare TS ($-1 \leq TS \leq 1$) with 0.7, independent ft on TS provided correct formula used, or on CV 0.600 In context, not too positive. FT on TS only SC: 0.600 (2-tailed): B0 M1A0	
		Total	10		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
2			$r = 0.686(41)$ $H_0: \rho = 0, H_1: \rho > 0$ where ρ is the population correlation coefficient $r_{crit} = 0.6581$ Reject H_0 . There is significant evidence of positive correlation between the taster's estimates and the actual price.	M1 (AO3.4) A1 (AO1.1) B1 (AO1.1) B1 (AO2.5) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [7]	Evidence for correct method (e.g. correct value) Correct value B1 if one error or if ρ not defined as population parameter Seen. Allow 0.658 Correct first conclusion Contextualised, not too definite. FT on their r but not their CV	
			Total	7		



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Question			Answer/Indicative content	Marks	Part marks and guidance		
3	a	i	The points do not lie very close to a straight line	B1 (AO1.1) [1]	Or equivalent. Must refer to diagram, not just to "correlation"	Ignore extras unless wrong	
		ii	$H_0: \rho = 0$, $H_1: \rho > 0$, where ρ is the population pmcc between prices in 1972 and prices in 2018 $0.381 < 0.4973$ Do not reject H_0. There is insufficient evidence of (positive) correlation between prices in the two years. Insufficient evidence to reject H_0. No correlation between ... M1A1 (bod) Wrong first conclusion, correct interpretation: MOAO Hypotheses wrong way round maximum M1M1	B2 (AO1.1 2.5) M1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [5]	One error, e.g. ρ not defined, B1 (but allow "population" not stated) $H_0: r = 0$, $H_1: r > 0$: same scheme, but B2 needs "population" pmcc H_1: positive correlation: B1 Compare with 0.497(3) Correct first conclusion, needs like-with-like In context, not too definite	H_0: no correlation, FT on CV 0.5760 only	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	b		0.650	B2 (AO3.1a 1.1) [2]	Full marks for correct answer by any method	SC: if B0 allow B1 for any 3 of 8.85, 46.35, 8.8725, 241.7331, 43.153	
			Total	8			
4	a		$H_0: \rho_s = 0, H_1: \rho_s > 0$, where ρ_s is the population rank correlation coefficient Rank 5 4 3 6 1 2 s 5 1 2 6 4 3 $\Sigma d^2 = 20$ $r_s = 1 - \frac{6 \times 20}{6 \times 35}$ = 3/7 or 0.42857... < 0.9429 Do not reject H_0 Insufficient evidence of association between ranking given by the two categories	B1 (AO1.1) B1 (AO1.1) M1 (AO1.1) A1 (AO1.1) B1 (AO1.1) M1ft (AO1.1) A1ft (AO2.2b) [7]	Allow ρ_s not defined; allow ρ. Allow: H_0: no association between rankings. H_1: positive association (but not H_1: association) Exact or awrt 0.429 FT on their Σd^2 only		
	b		Not dependent on any distributional assumptions	B1 (AO1.2) [1]	Oe (cf. Specification, 5.08f)		
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance	
5	a		-0.954 BC	B2 (AO1.1) (AO1.1) [2]	SC: If B0, give B1 if two of 7.04, 29.0[4], -13.6[4] (or 35.2, 145[.2], -68.2) seen	
	b		Points lie close to a straight line Line has negative gradient	B1 (AO2.2b) B1 (AO1.1) [2]	Must refer to line, not just "negative correlation"	
	c		No, it will be the same as $x \rightarrow a$ is a linear transformation	B1 (AO2.2a) [1]	OE. Either "same" with correct reason, or "disagree" with correct reason. Allow any clear valid technical term	
			Total	5		



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Question		Answer/Indicative content	Marks	Part marks and guidance		
6	a	H_0 : no association between orders in races of this type; H_1 : positive association between orders	B1(AO1.1) [1]	Or clear equivalent, e.g. agreement, independent, or $H_0: \rho_s = 0$, $H_1: \rho_s > 0$ Not "times"	Allow ρ but not r or r_s unless "population" explicit	
<u>Examiner's Comments</u> Again, many candidates had not appreciated that hypotheses should be stated in terms of a population parameter (ρ (or preferably ρ_s) as opposed to r), as was flagged in the specimen material. On this particular occasion credit was given for verbal explanations.						



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	$1 - \frac{6\Sigma d^2}{5 \times 24} \geq 0.9$ $\Rightarrow \Sigma d^2 \leq 2$ Largest possible value is 2	M1(AO3.1a) M1(AO1.1a) A1(AO1.1) [3]	Use correct formula and any tabular value of r_s (not 0.05 or from r) Solve for Σd^2 , needs correct formula Correct conclusion, www but allow = throughout	Allow > or = $\leq$ then $\Sigma d^2 \geq 2$: M1M1A0 Allow from 0.05 Allow $\Sigma d^2 < 2$ so $\Sigma d^2 = 1$ if working correct	

Examiner's Comments

Most candidates correctly obtained 0.9 from the tables (although a few used the pmcc value), and most also used the correct formula for Spearman's rank correlation coefficient. The use of inequalities was very poor, however. Many used = throughout and then often went on to assume that Σd^2 was greater than their value, instead of equal to (or less than) it.

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Question		Answer/Indicative content	Marks	Part marks and guidance		
	c	$\Sigma d^2 = 2 \Rightarrow d = 0, 0, 0, 1, -1$ in some order E.g. ABCED or 12354 (allow more than one correct answer) SC two-tailed (<u>must be from</u> <u>"$H_1: \rho_s \neq 0$" only</u>): B0; $1 -$ $6\Sigma d^2 / 120 \geq 1.0$, $\Sigma d^2 = 0$ M1A1; "same order only" B1, total 3/7 SC wrong Σd^2 : in (c) allow M1M1 for showing any order giving $\Sigma d^2 < \text{their (b)}$	M1ft(AO3 .1b) M1ft(AO2 .2a) A1(AO2. 2b) [3]	Turn $\Sigma d^2 \leq$ their 2 into possible values of d Deduce possible order Any order with two consecutiv e runners i nterchange d, allow from $\Sigma d^2 =$ 3 but no other values	SC: $\Sigma d^2 =$ 1, d = 00001 M1 1 explicitly impossible A1 Same order only A1 Else ignore ABCDE if seen	
		Total	7	<u>Examiner's Comments</u> As above, some candidates found rank orders that gave totals bigger than their answer to part (b), rather than less than it. However, most who used $\Sigma d^2 = 2$ gave one of the possible correct orders.		



Question			Answer/Indicative content	Marks	Part marks and guidance	
7	a		0.8392	B1(AO1.1) [1]	Awt 0.839 $S_{xx} = 1.7449...$ $S_{yy} = 41.2...$ $S_{xy} = 7.116...$ <u>Examiner's Comments</u> For most candidates this was a very straightforward start to the paper.	
	b		$y = -1.180 + 4.0781x$	B1(AO1.1) B1(AO1.1) [2]	Both coeffs, awrt -1.18 and 4.08 Letters correct, needs 1 correct coefficient <u>Examiner's Comments</u> Correct answers were very common here, although a few candidates used their product-moment correlation coefficient as a. The variable letters x and y were almost always correctly given.	
	c		Value of PMCC suggests that there is strong correlation, or 0.75 <u>shown</u> close to mean 0.399	B1(AO3.5a) [1]	E.g. "r high so points lie close to line". "r is high" alone is enough. No wrong extras Not "0.75 is close to mean", unless properly justified, e.g. SD (= 0.264) calculated <u>Examiner's Comments</u> Many candidates produced unnecessarily complicated answers. There are only two standard criteria to consider for the reliability of an estimate, correlation and	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					whether the given data value (here 0.75) is in the data range. As candidates should have got in Question 1(a) a value of r that is high, there was no reason for them to respond with an answer any more complicated than 'r is close to 1 so there is strong correlation'. Those who attempted to use the other condition needed to justify that 0.75 was likely to be within the range of the data (for instance by finding both the sample mean (0.399) and the standard deviation), but this cannot be known with certainty from the information given. Exemplar 1 No) The estimate will be reliable when $n > 0.75$ because n is a large number of 25 so a large sample is used to find the regression line so anomalies will not affect it too much. This is a typical irrelevant answer. Exemplar 2 No) The average value of r is 0.399 and $r = 0.75$ is close to this so the data range is not too far. This assertion is not valid without further calculation, for example of the standard deviation. The deleted words make no difference.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
	d		Whether $x = 0.75$ is within the data range	B1(AO 3.5b) [1]	E.g. “maximum and minimum values of x ”; not “all data points”. No wrong extras	Or clear reference to interpolation. NB: 95% CI for x is (-0.156, 0.954)
					<u>Examiner's Comments</u> Following from the comments on part (c), it should be clear that all that was needed here was ‘the range of values of x ’. It is not necessary to know all the data values and those who said that they need to know the correct corresponding value of y were missing the point. Some said that the data had to be bivariate normal, or elliptically distributed, but this is not necessary unless a formal hypothesis test is being carried out. About half of the candidates picked up the mark here.	
			Total	5		

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Question		Answer/Indicative content	Marks	Part marks and guidance		
8	a	$H_0: \rho = 0$, $H_1: \rho \neq 0$, where ρ is population pmcc $0.701 > 0.4973$ Reject H_0. There is significant evidence of association between the marks on the two papers	B1(AO2.5) B1(AO1.1a) M1ft(AO1.1) A1(AO2.2b) [4]	Must use symbols. Allow no definition of letter if ρ used Correct CV stated, allow 0.497 FT on wrong CV Not FT. Needs context, and not too definite.	Not "H_0 : there is no assoc'n, H_1: there is association" Not <u>There is</u> association ..."	<u>Examiner's Comments</u> The mark scheme of the Specimen Paper made it clear that hypotheses had to be stated in terms of the population parameter ρ, and that verbal statements such as 'H_0: there is no association between the marks on the papers' would not gain credit. Candidates should also state clearly that ρ is the <u>population correlation coefficient</u>. Exemplar 3 indicates an acceptable explanation. Many chose the wrong critical value, typically 0.4259 (the one-tailed value) instead of the correct 0.4973. However, comparisons and conclusions were often stated clearly and carefully. As usual, conclusions to hypothesis tests need to be contextualised (here requiring phrases like 'association <u>between</u>



Question			Answer/Indicative content	Marks	Part marks and guidance	
					<u>candidates' marks')</u> and with acknowledgement of uncertainly (so not 'there is association ...' but 'there is <u>significant evidence of association ...</u> '). Some gave the conclusion as ' <u>insufficient evidence of <u>no</u> association</u> ', which is wrong.	
	b	i	-0.534	B2(AO1.1a) (AO1.1) [2]	SC: if B0, give B1 for two of 1440, 2066, -921 and $S_{xy}/\sqrt{(S_{xx}S_{yy})}$	-0.53: B1
					<u>Examiner's Comments</u> Almost always right. Most candidates showed a good quantity of working.	
		ii	6 candidates did very well or very badly on both papers; middle 10 tended to do badly on one paper and well on the other	B1(AO2.4) [1]	Correct inference about scores oe, not "correlation / associatio n/value of r". Not "outliers" or "anomalies".	Allow inference for one group only, provided it is clearly for only one group & any ref to other group is not wrong
					<u>Examiner's Comments</u> The wording of this question required an inference about the <u>scores</u> of the candidates, not about correlation or association, so that answers such as Exemplar 3 did not gain the mark. Nor does it make sense to reject 6 candidates out of 16 as outliers. Good	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					answers showed good statistical literacy, drawing inferences from the information, for example 'the best candidates did well on both papers, the worst did badly on both the others tended to do worse on one paper the better they did on the other'. Some answers referred to 'a large range of results'; whether or not the use of the word 'range' here is intended to be technical, this cannot be inferred from the values of r. Answers such as 'the results are very varied' revealed a limited understanding of the concepts. Some candidates were unable to engage with the question and either left it blank or regurgitated irrelevant answers such as 'the data must be bivariate normal' or 'correlation is not causality'. Exemplar 3 5(0)(b) A strong positive correlation for the whole class, but a negative correlation for the middle 10 students. This answer refers merely to correlation and does not draw any inferences about the scores.	
			Total	7		

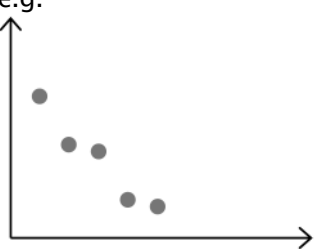
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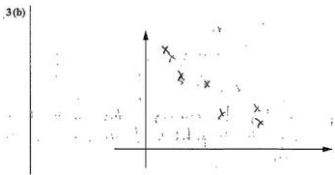
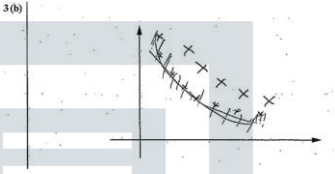


Question		Answer/Indicative content	Marks	Part marks and guidance		
9	a	Minimum possible non-zero value of Σd^2... ... is 2 $1 - \frac{6 \times 2}{9 \times 80}$ $\frac{59}{60}$ or 0.983(33...)	M1(AO3.1a) A1(AO2.2a) M1(AO1.1a) A1(AO1.1) [4]	Find minimum Σd^2 2 stated or used Use correct formula Answer, exact or 0.983 or better	Allow for $\Sigma d^2 = 1$ Allow $1 - \frac{6 \Sigma d^2}{9 \times 80}$ <u>us</u> w <u>ed</u>	<u>Examiner's Comments</u> Many saw quickly that the minimum value of Σd^2 had to be 2, and then quickly got the correct answer. Others spent time trying to use inequalities to obtain $\Sigma d^2 > 0$, after which they still had to find the number 2. Quite a lot of candidates thought that the minimum value of Σd^2 could be 1.



Question		Answer/Indicative content	Marks	Part marks and guidance		
	b	e.g. 	M1(AO1.2) A1(AO1.2) [2]	Points strictly decreasing Not in straight line, no errors SC: Curve, no points: M1	Ignore lines drawn < 5 points: M1A0. > 5 points: M1A1 Needn't be in first quadrant	<u>Examiner's Comments</u> Many found it easy to draw such a diagram (although, even though a number of diagrams had more than 5 points, which did not matter). But others did not seem to have the proper concept, and suggested answers included scatter graphs with points either side of a line, points on a curve with a turning point, or graphs with positive gradient. This seemed to be an example of a question that identified candidates who could answer questions only if they had seen them before. Some thought that '5 pairs' meant 10 points. Those candidates who used dots rather than crosses, or who (as mentioned before) changed their answer using an eraser or correcting fluid, caused problems for examiners. The scanning process can often make erased objects as visible as non-erased ones. The issues can be seen from the following exemplars.



Question			Answer/Indicative content	Marks	Part marks and guidance	
					Exemplar 1 	
					Here it is hard for the examiner to see what has been deleted. The examiner will give benefit of doubt where possible, but it is not always possible. Exemplar 2 	
			Total	6		

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Question			Answer/Indicative content	Marks	Part marks and guidance		
10		a	$r = -0.55397$	B2 (AO 1.1) (AO 1.1) [2]	art -0.554	SC: If B0, allow M1 for correct method with S_{xx} , S_{yy} , S_{xy} etc	
		b	None	B1 (AO 1.2) [1]	"Unchanged", same value written out, etc		
		c	Increasing birth rate is associated with lower life expectancy	B1 (AO 2.2b) [1]	oe		
		d	$H_0: \rho = 0$ $H_1: \rho \neq 0$ where ρ is the population pmcc $CV \pm 0.5324$ $-0.55397 < -0.5324$ so reject H_0 There is significant evidence of association between birth rate and life expectancy	B2 (AO 1.1) (AO 2.5) B1 (AO 1.1) M1ft (AO 2.2b) A1ft (AO 1.1) [5]	One error or omission, e.g. no symbol, <, ρ not defined: B1 Ignore sign for this mark Correct first conclusion, needs consistent sign In context, not too definite [not: there is association ...]	ft on r only if consistent Allow in terms of ρ only if contextualised	
		e	No as 3.9 is outside the data range	B1 (AO 2.3) [1]	Or Zimbabwe different, etc		
			Total	10			



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	-0.85462 [$S_{xx} = 1.7656$, $S_{yy} = 5.7096$, $S_{xy} = -2.4644$ or 8.828, 23.548, -12.322]	B2 (AO 1.1) (AO 1.1) [2]	BC -0.855 or better	SC: If B0 give M1 for correct use of Ss	
		b	Points lie (quite) close to a straight line with negative gradient	B1 (AO 2.2b) [1]	Both nearness and sign needed		
		c	$H_0: \rho = 0$, $H_1: \rho \neq 0$, where ρ is the population product moment correlation coefficient CV = (-) 0.8783 -0.855 > -0.8783 so do not reject H_0 Insufficient evidence of association	B2 (AO 1.1a) (AO 2.5) B1 (AO 1.1) M1 (AO 2.2b) A1 (AO 3.5a) [5]	ρ not properly defined in terms of population: B1 Correct CV Comparison and first conclusion Not too definite, e.g. not "there is no association"	No context needed	
		d	-1	B1 (AO 2.2a) [1]	Not "nearly -1"		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		e	(i Neither is changed) (i r is changed but r_s is not i)	B1 (AO 2.2a) [1] B1 (AO 2.2a) [1]			
			Total	11			



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Question			Answer/Indicative content	Marks	Part marks and guidance	
12		a	0.4(0 BC 0)	B2(AO1.1) (AO1.1b) [2]	SC: if B0, give SC B1 for two of $\textcircled{1}_{xx} = 12500,$ $\textcircled{1}_{yy} = 1600,$ $\textcircled{1}_{xy} = 1790$ and $\textcircled{1}_{xy} / \sqrt{(\textcircled{1}_{xx} \textcircled{1}_{yy})}$ Also allow SC B1 for equivalent methods using Covariance & SDs	
		b	Data needs to have a bivariate normal distribution	B1(AO1.2) [1]	Needs “bivariate normal” or clear equivalent. Not just “both normally distributed” Allow “scatter diagram forms ellipse” <u>Examiner’s Comments</u> This request seemed unfamiliar to a majority of candidates. Either of the answers “the data must have a bivariate normal distribution” or “the points should form an approximate ellipse on a scatter diagram” was acceptable. If this assumption does not hold, the hypothesis test is not valid.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	H_0: higher maths scores are not associated with higher BCF grading; H_1: positively associated CV 0.3783 $0.400 > 0.3783$ so reject H_0 Significant evidence that higher maths scores are associated with higher BCF grading	B1(AO2.5) B1(AO1.1b) M1ft(AO2.2b) A1ft(AO3.5a) [4]	Needs context and clearly one-tailed OR p used and defined. Not "evidence that ..." Allow 0.378 Reject/do not reject H_0 Contextualised, not too definite Needn't say "positive" if H_1 OK SC 2-tail: B0; 0.4438, or 0.3783 B1; then M1A0 <u>Examiner's Comments</u> Most candidates found the correct critical value. They also gave their conclusions in context and with acknowledgement of the uncertainty involved. A good conclusion might be, "Reject H_0. There is significant evidence that chess players with higher BCF ratings are better at mathematics." Problems arose with stating the hypotheses. Good statements were of one of	$H_0: \rho = 0$, $H_1: \rho > 0$ where ρ is population pmcc (not r) FT on their r, but not CV Not "scores are associated...". FT on their r <u>only</u>

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Question			Answer/Indicative content	Marks	Part marks and guidance	
					two forms: $H_0: \rho = 0$, $H_1: \rho > 0$, where ρ is the population product moment correlation coefficient OR H_0: there is no association between BCF rating and mathematical ability. H_1: there is positive association between BCF rating and mathematical ability. A definition of ρ is crucial, including mention of the word “population” It was also important to include the word “positive” in H_1, since its absence would make their test 2-tailed instead of 1-tailed. A number of candidates stated their hypotheses the wrong way round: “H_0: there is positive association ...”. Some wrote: H_0: there is no evidence of association ...”, but “evidence” refers only to the conclusion. The hypotheses refer to the actual underlying distribution, where either there is positive association or there is not.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	It makes no difference as this is a linear transformation	B1(AO2.2a) [1]	Need both “unchanged” oe and reason, need “linear” or exact equivalent	“oe” includes “their 0.4”	
					<u>Examiner's Comments</u> A large number of candidates correctly said that there would be no effect because linear coding does not affect the value of the pmcc. Some candidates said that ELO ratings have a wider geographical validity than BCF ratings, but this is irrelevant to the context of the experiment.		
			Total	8			



Question			Answer/Indicative content	Marks	Part marks and guidance		
13		a	H_0: no association between rankings H_1: positive association Rankings 1 2 3 4 5 6 7 2 4 1 3 5 6 7 $r_s = 1 - \frac{6 \times 10}{7 \times 48} = 0.8214...$ $0.8214 < 0.8929$ Do not reject H_0. Insufficient evidence of agreement between judges' rankings.	B1 (AO 2.5) M1 (AO 1.1a) M1 (AO 3.1a) A1 (AO 1.1) B1 (AO 1.1a) M1ft (AO 2.2b) A1ft (AO 3.5a) [7]	Attempt to rank $O \frac{23}{28}$ Correct comparison FT on their TS or on 0.9286 Contextualized, not too definite	Allow rankings in opposite order NOT "there is agreement"	
		b	(i) $1/7!$ $= \frac{1}{5040}$ or 0.000198... (ii) $1 - \frac{6 \sum d^2}{7 \times 48} = \frac{55}{56} \Rightarrow \sum d^2 = 1$ Impossible as 1 incorrect ranking implies another	M1 (AO 3.1a) A1 (AO 1.1a) [2] M1 (AO 2.1) A1 (AO 2.2a) [2]	Exact or awrt 0.000198 Find $\sum d^2$ Justify impossible		
			Total	11			



Question			Answer/Indicative content	Marks	Part marks and guidance	
14		a	$1 - \frac{6\Sigma d^2}{7 \times 48} = \frac{27}{28} \Rightarrow \Sigma d^2 = 2$ $\Rightarrow d_i = 0, 0, 0, 0, 0, 1, 1$ in some order 6 possibilities $\frac{1+6}{7!}$ $= \frac{1}{720}$ or 0.0013888...	M1(AO 3.1b) A1(AO 1.1) A1(AO 2.2a) B1(AO 1.1) M1(AO 1.1a) A1(AO 1.1a) A1(AO 2.1) [7]	$1 - \frac{6\Sigma d^2}{7 \times 48} = \frac{27}{28}$; correct Σd^2 Stated or implied Seen or implied Divide by 7! 1 included, www Www, exact or art 0.00139	
		b	0.1388...%	B1(AO 3.1b) [1]	Allow 0.00139 etc, or same as (a) Not 0.00139%	
			Total	8		

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Question			Answer/Indicative content	Marks	Part marks and guidance											
15		a	Expected frequencies 20, 12, 8; 10, 6, 4 One expected frequency is less than 5	B1*(AO 1.1) depB1 (AO 2.3) [2]												
		b	Mathematics and Physics are the more closely related subjects	B1(AO 2.4) [1]												
		c	H ₀ : No association between subject and year H ₁ : there is association between subject & year Combine cells to obtain <table><tr><th colspan="2">Observed</th><th colspan="2">Expected</th></tr><tr><td>24</td><td>16</td><td>28</td><td>12</td></tr><tr><td>18</td><td>2</td><td>14</td><td>6</td></tr></table> $\sum \frac{(O - E - 0.5)^2}{E}$ = 0.4735 + 1.0208 + 0.875 + 2.042 = 4.375 > 2.706 Reject H ₀ . There is significant evidence of association between choice of subject and year of entry	Observed		Expected		24	16	28	12	18	2	14	6	B1(AO 1.1) <
Observed		Expected														
24	16	28	12													
18	2	14	6													



Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	English / Year 12 entry	B1(AO 2.2a) [1]	FT on Physics & English makes no sense here		
		e	Significantly fewer students who enter in Year 12 choose English than would be expected if there were no association.	B1 FT (AO 3.2a) [1]			
			Total	11			
16			$P(R=7) > P(R=6) \Rightarrow e^{-\lambda} \frac{\lambda^7}{7!} > e^{-\lambda} \frac{\lambda^6}{6!}$ $6! \lambda^7 > 7! \lambda^6$ $\Rightarrow \lambda > 7$ $P(R=7) > P(R=8) \Rightarrow e^{-\lambda} \frac{\lambda^7}{7!} > e^{-\lambda} \frac{\lambda^8}{8!}$ $\Rightarrow \lambda < 7$ $7 < \lambda < 8$	M1*(AO 3.1a) A1(AO 1.1) M1(AO 1.1a) A1(AO 1.1) depM1 (AO 2.1) A1(AO 1.1) A1(AO 2.2a) [7]	One correct Poisson prob seen Correct inequality Reduce to linear equation in λ One correct limit Second inequality Second correct limit Correct double inequality	Allow = Allow = Allow = only if < later	
			Total	7			



Question			Answer/Indicative content	Marks	Part marks and guidance		
17		a	t is a controlled variable	B1(AO 2.5) [1]			
		b	0.987	B1(AO 1.2) [1]	Or “unchanged”		
		c	1	B1(AO 2.2a) [1]			
		d	Points lie close to a straight line	B1(AO 2.2b) [1]	Not “positive gradient” – that information is given by r_s		
			Total	4			



Question			Answer/Indicative content	Marks	Part marks and guidance		
18		a	N is likely to be dependent on T because more ice creams are likely to be sold in hot weather	B1(AO2.5) [1]	oe, any plausible reason that shows understanding of concept		
		b	$r = 0.819$	B2(AO1.1b)(AO1.2) [2]	BC: awrt 0.819 Correct answer, no working: B2	SC: answer incorrect, correct working seen: B1	
		c	No change, r unchanged by linear coding	B1(AO2.3) [1]	Both "no change" oe and "linear" oe		
		d	$n = -15.6 + 1.81t$	B1ft(AO1.1b)depB1(AO1.1b) [2]	BC, Numbers correct, to 3 sf Whole equation correct		
		e	Vertical line drawn between line and a point Minimise sum of squares of these lengths	M1*(AO2.4)depE1(AO2.4) [2]	At least one line drawn This or equivalent stated		
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Question			Answer/Indicative content	Marks	Part marks and guidance		
19		a	Gives $-\frac{37}{35}$ which is < -1	M1(AO2.3) A1(AO2.2a) [2]	Aim to establish in consistency Fully correct argument	Or: $r = -1$ $\Rightarrow \max \Sigma d^2 = 70$	
		b	$1 - \frac{6\Sigma d^2}{6(6^2 - 1)} = \frac{29}{35}$ $\Rightarrow \Sigma d^2 = 6$	M1(AO1.1b) A1(AO2.2a) [2]	Use formula and solve		
		c	H_0 : No association between critics' rankings H_1 : Some association CV 0.886 $0.829 < 0.886$ so do not reject H_0 Insufficient evidence of association in the rankings	B1(AO2.5) B1(AO1.1b) M1(AO1.1b) A1(AO2.2b) [4]	Or $0.829 < 0.8857$ In context, acknowledge uncertainty	Not, e.g. "there is no association between the rankings"	
		d	Not necessarily as testing different things (association / correlation)	B1(AO2.4) [1]	Allow, e.g.: "numerical ratings likely to be arbitrary" "even if raw data are not linearly correlated, ranks might be identical", etc		
			Total	9			



Question			Answer/Indicative content	Marks	Part marks and guidance		
20		a	$y = 157x - 988$	B1(AO1.1) B1(AO3.3) [2]	Either coefficient correct to 3 s.f. All correct including letters	BC [$y = 157.11x - 988.15$]	
		b	No; r is unaffected by linear coding	E1(AO2.3) [1]	Needs both "no" and "linear" seen, and no wrong or irrelevant statements		
		c	(i) $H_0: \rho = 0$ $H_1: \rho \neq 0$ where ρ is the population correlation Coefficient CV 0.5494 $0.9586 > 0.5494$ Reject H_0 Significant evidence of correlation between consumption of mozzarella cheese and number of civil engineering doctorates awarded.	B2(AO1.1) (AO2.5) B1(AO3.4) M1(AO1.1) M1(AO1.1) A1(AO2.2b) [6]	Hypotheses correctly stated B1 if no symbol used Correct critical value seen Explicit comparison Correct first conclusion Contextualised, acknowledged uncertainty		



Question			Answer/Indicative content	Marks	Part marks and guidance															
		c	(ii) No, correlation does not imply causality	E1(AO3.5b) [1]	Or: "there may be another factor that influences both"															
			Total	10																
21		a	Rankings <table border="1"> <tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr> <tr><td>1</td><td>3</td><td>2</td><td>4</td><td>7</td><td>5</td><td>6</td></tr> </table> $\Sigma d^2 = 8$ $r_s = 1 - \frac{6 \times 8}{7 \times 48} = \frac{6}{7}$ or 0.857	1	2	3	4	5	6	7	1	3	2	4	7	5	6	*M1(AO1.1a) A1(AO1.1) dep*M1(AO1.1) A1(AO1.1) [4]	Rank both, can be reverse order $\Sigma d^2 = 8$ BC Use of correct formula awrt 0.857	
1	2	3	4	5	6	7														
1	3	2	4	7	5	6														
		b	Strong agreement/association in rankings/views	B1(AO2.2b) [1]	Not strong correlation/relationship															
			Total	5																



Question			Answer/Indicative content	Marks	Part marks and guidance	
22		i	$\begin{array}{ccccc} 5 & 2 & 4 & 1 & 3 \\ 3 & 4 & 5 & 2 & 1 \end{array}$ or A B C D E (grades) $\begin{array}{ccccc} & & & & \\ & & & & \end{array}$	M1	Attempt ranks	
		i		A1	Correct ranks; allow both sets reversed Can be implied by eg $\Sigma d^2 = 14$	One set reversed: A0 Use PMCC on ranks: 1 st M1A1 as main scheme then: $\Sigma x = \Sigma y = 15$ $\Sigma x^2 = \Sigma y^2 = 55$ $\Sigma xy = 48$
		i	$\frac{d^2}{\Sigma d^2} \begin{array}{ccccc} 4 & 4 & 1 & 1 & 4 \end{array}$ $(= 14)$	M1	Attempt Σd^2 dep 1 st M1	$S_{xx} = S_{yy} = 10$ $S_{xy} = 3$ allow one arith error M1
		i	$1 - \frac{6 \times 14}{5 \times (5^2 - 1)}$	M1	ft Σd^2 dep 1 st M1 If one set reversed, $r_s = -0.3$	$r = 3 / \sqrt{(10 \times 10)}$ allow one arith error M1

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	= 0.3 oe	A1	M1A0M1M1A0 <u>Examiner's Comments</u> Some candidates ranked the marks in the order "smallest first" (presumably working on auto-pilot), and the grades in alphabetical order. This means that one set of ranks is in reverse order and thus leads to the incorrect answer of -0.3 instead of 0.3. Others just made errors in their ranking. A few ranked the students 1, 2, 3, 4, 5 instead of the marks. As usual, a few used the original marks together with ranks for the grades, obtaining impossible values for r_s . Incorrect versions of the formula, most commonly $\frac{6 \times \sum d^2}{n(n^2-1)}$ and $\frac{1-6 \times \sum d^2}{n(n^2-1)}$, were sometimes seen. A few candidates used the formula for the PMCC on the ranks, which is a valid method, but is long and gives more opportunities for arithmetical errors than Spearman's formula.	= 0.3 A1
		ii	$\sum d^2 = 8$ or '2 the same and 2 differ by 2'	M1	May be implied	Allow $d^2 = 8$ or similar
		ii	1 4 3 2	A1	<u>Examiner's Comments</u> This unusual question was well answered by many. A few just reversed Judge 1's ranks: 4, 3, 2, 1.	



Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	7	



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Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	$S_{xx} = 503.45 - \frac{70.3^2}{10}$ (= 9.241) $S_{yy} = 103.94 - \frac{30.8^2}{10}$ (= 9.076)			
		i	$S_{xy} = 211.9 - \frac{70.3 \times 30.8}{10}$ (= -4.624)	M1	Correct sub in any correct ① formula	
		i	= -0.5049.... or -0.505 (3 sfs)	M1	Correct sub in any correct r formula	Must be correct sub in all ①'s & r but not nec'y accurate
		i	= -0.5049.... or -0.505 (3 sfs)	A1	Correct ans with no wking: M1M1A1	
					<u>Examiner's Comments</u> This part was well answered. A few candidates made errors such as those mentioned above. Some candidates used the statistical functions on their calculator and gave an answer with no working. As noted above, this is a risky strategy, resulting in either full marks or no marks. Candidates who wish to use this method should be advised to carry out the whole calculation a second time, as a check. The safest approach is to show the substitutions into the relevant formulae for ① _{xy} etc and for b and r.	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Correlation (of UR & CPI) does not imply causation or r not close to -1	B1	Both (UR & CPI) may depend on another factor or r small or poor corr'n or ignore all else <u>Examiner's Comments</u> A good number of candidates answered correctly either by pointing out that correlation does not imply causation or that other factors may be involved. Many, however, stated (wrongly) that the reason why the given statement is wrong is that the data in the table show that the progression is sometimes up and sometimes down.	Allow One may depend on another factor Allow without context NOT eg UR is independent NOT eg Only for the given years NOT eg Only for certain months
		iii	$b' = \frac{s_{xy}}{s_{yy}} = \frac{-4.624}{9.076} (= -\frac{1156}{2269} \text{ or } -0.50948)$	M1	ft their ①'s	If y on x: $b = \frac{s_{xy}}{s_{xx}} = \frac{-4.624}{9.241} (= -0.500)$ M1
		iii	$x - \frac{70.3}{10} = -\frac{1156}{2269} (y - \frac{30.8}{10})$	M1	or $a' = -\frac{1156}{2269} \times (-\frac{30.8}{10}) + \frac{70.3}{10}$	$y - 3.08 = -0.500 \times (x - 7.03)$ or $a = 3.08 + 0.5 \times 7.03$ M1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$x = -0.51y + 8.6$ (2 sfs) or $x = -\frac{1156}{2269}y + 8.6$	A1	NB use b' (= -0.509), not r (= -0.5049) <u>Examiner's Comments</u> Some candidates misread the question and found the equation of the (more usual) y on x line. However, the majority read the question correctly and attempted to find the equation of the x on y line. But many of these failed to be consistent in their intention. Some found the gradient (b') correctly but then reverted to the method for y on x , attempting $a' = \bar{y} - b' \times \bar{x}$, which is incorrect. Still others found b' and a' correctly but wrote their final answer as $y = b'x + a'$ instead of $x = b'y + a'$	$y = -0.50x + 6.6$ A0
		iv	$x = -0.509 \times 4.0 + 8.60$	M1	Allow sub $y = 0.04$ for M1 only	If y on x found in (a)
		iv	= 6.56 (3 sf) or 6.6 (2 sf)	A1ft	ft their equn; ans to 2 sf <u>Examiner's Comments</u> This was well answered, with follow-through from incorrect equations allowed. A few candidates substituted for x instead of y .	$4.0 = -0.500x + 6.60$ M1 $x = 5.2$ (2 sf) A1ft
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$S_{xx} = 8700000 - \frac{7000^2}{6} \quad (= 533333)$ $S_{xy} = 509900 - \frac{7000 \times 456}{6} \quad (= -22100)$	M1	Correct subst'n in any correct ① formula	
		i	$b = -\frac{22100}{533333} \text{ or } -\frac{663}{16000} \quad (= -0.0414)$	M1	Correct subst'n in any correct b formula from two correct ① formulae	
		i	$y - \frac{456}{6} = "-0.0414"(x - \frac{7000}{6})$	M1	ft their b except if using r	or $a = \frac{456}{6} - ("-0.0414") \times \frac{7000}{6}$ oe ft "b"
		i	$y = -0.0414x + 124 \text{ (3 sf)}$	A1	or $y = -\frac{663}{16000}x + \frac{3979}{32}$ or $y = -0.041x + 124$ Examiner's Comments This part was answered very well on the whole. A few candidates made a sign error when substituting b (which is negative) in order to find a. Some simply lost the minus sign in b. Some found b and stopped. Others found r instead of what was asked. A few found r and then used this as their value of b.	Allow $y = -0.04x + 124$ if -0.041... seen
	ii	70 to 72		B1	or 71 per thousand, NOT 71000 Examiner's Comments Many candidates ignored the instruction to "use the regression line in the diagram" and used their equation from (i). A few candidates misinterpreted the situation, giving an answer such as 71 000.	No ft from (i) Ignore method



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Extrapolation oe	B1	Allow "2400 is beyond graph" } "Not shown on the graph" or } 1 st B1 only "Line drops low, or below 0" } "Outlier" }	"Line only allows for countries poorer than Nigeria" 1 st B1 Allow "Value for Nigeria is -ve 1 st B1
		iii	Corr'n not high or small sample	B1	Poor corr'n oe, or pts not close to line oe 2 nd B1 Examiner's Comments Most candidates gave one correct answer, using the word "extrapolation" or some equivalent wording. However, many either gave no second reason or gave one that was, in effect, equivalent to their first reason (eg "The IMR will become negative"). Some candidates gave the valid second reason, namely that the diagram does not show good linear correlation.	NOT "Other factors may apply" oe Ignore all else
		iv	$S_{xx} = 8700000 + 1300^2 - \frac{(7000+1300)^2}{7}$		or $10390000 - \frac{(8300)^2}{7} = \frac{3840000}{7}$ or 548571	
		iv	$S_{yy} = 36262 + 96^2 - \frac{(456+96)^2}{7}$	M1	or $45478 - \frac{552^2}{7} = \frac{13642}{7}$ or 1948.86	Correct sub in any correct ① formula M1 Correct value of any ① seen or implied by r A1
		iv	$S_{xy} = 509900 + 1300 \times 96 - \frac{8300 \times 552}{7}$	A1	or $634700 - \frac{8300 \times 552}{7} = -\frac{138700}{7}$ or -19814.3	
		iv	$r = \frac{-19814.3}{\sqrt{548571 \times 1948.86}}$	M1	Correct subst'n in any correct r formula from 3 correct subs in 3 correct ① formulae, ie all correct method	SC If n = 6, but otherwise correct allow M1A0M1A0 (ans r = -0.574, must see wking)



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= -0.606 (3 sf)	A1	Examiner's Comments This part was answered well by most candidates. A few used the original totals, just changing the value of n from 6 to 7. Others made the opposite error, finding new totals, but with $n = 6$. Sensibly, most wrote down their new totals such as Σxy, but some were incorrect and, without any indication of method, these lost a method mark.	
		v	No effect oe	B1	Stay the same oe Allow just "No" Examiner's Comments A few candidates thought that r would decrease because the values used in the formula would decrease, but most stated correctly that r would be unchanged.	Ignore all else
			Total	12		

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Question			Answer/Indicative content	Marks	Part marks and guidance	
25		i	$\Sigma d^2 = n$ seen or implied	M1		Trial method:
		i	$1 - \frac{6 \times \text{anything}}{n(n^2-1)} = \frac{63}{65}$ or $\frac{6 \times \text{anything}}{n(n^2-1)} = \frac{2}{65}$	M1	eg $1 - \frac{6 \times \Sigma d^2}{n(n^2-1)}$ or $1 - \frac{6 \times n^2}{n(n^2-1)}$ or $1 - \frac{6 \times 1^n}{n(n^2-1)}$ or $1 - \frac{6 \times 6^2}{n(n^2-1)} = \frac{63}{65}$	$\Sigma d^2 = 14$ M1 $1 - \frac{6 \times 14}{14(14^2-1)} = \frac{63}{65}$ A1 (0.969: A0)
		i	$\frac{6}{(n^2-1)} = \frac{2}{65}$ or eg $390 = 2(n^2-1)$	A1 depM2	Any correct eqn after cancelling n or take out factor of n; can be implied by $n = 14$	$\Rightarrow n = 14$ A1 Conclusion needed

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$n = 14$ NOT $n = \pm 14$	A1	But A0 if $n = 14$ clearly follows from incorrect working If no working or unclear working, but $n = 14$, M1M1A1A1 Examiner's Comments Most candidates correctly attempted to form an equation by equating the formula for r_5 to $\frac{63}{65}$. Many did not know what to do with the Σd^2. Some substituted $\Sigma d^2 = 6$ (or 8) by looking at the table. Others recognised $d = 1$ but thought that $\Sigma d^2 = 1$. Some understood that $d = 1$ for n pairs and while a few put $\Sigma d^2 = 1^n = 1$, many found the correct value of $\Sigma d^2 = n$. From then on the problem was one of algebra. Some cancelled the n immediately and went on to solve the equation correctly. Others did not cancel and found themselves faced with a cubic equation. Some cancelled at that stage although some did so incorrectly. Many algebraic errors crept in. A few candidates simply tried a few values of n and generally found the correct value in the end.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(a) $r = 1 \Rightarrow$ st line, hence true (or $r_s = 1$) oe Explanation essential Must state or imply "true"	B1	$r = 1 \Rightarrow y$ incr as x incr, so $r_s = 1$ oe Allow "True because perfect corr'n" or "True because $r = 1$ means pts ranked in order so $r_s = 1$ " " $r = 1$ means the ranks will agree" " $r = 1$ means all d's are 0, hence $r_s = 1 - 0 = 1$ " Examiner's Comments Many candidates answered this part correctly, often referring to a straight line or to perfect correlation. A few thought that the statement is false because r and r_s are independent of each other.	NOT "r incr so ranks incr" NOT " $r_s = r$ for ranks so true" NOT "True because strong corr'n"
		ii	(b) Diag, ≥ 3 pts, not on st line but with $x_{n+1} > x_n$ & $y_{n+1} > y_n$, Zig zag line or curve, moving up & right	B1	Ignore explan if correct diag given Ignore any st line drawn Allow numerical example for which $r \neq 1$ but $r_s = 1$. If expl'n contradicts diag, mark diag	
		ii	so r_s can still be 1	B1dep	For 2 nd B1 must state or imply "false"	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	eg "exponential curve gives $r \neq 1$ but $r_s = 1$ B1B1		Examiner's Comments Candidates struggled to give a convincing explanation here. Some unwisely ignored the instruction to "use a diagram" thus compounding their difficulties. A few clearly understood the point but drew a diagram in which one point was out of position, thus invalidating their argument. Others drew a diagram with $r \approx -1$ and $r_s = -1$. Some drew incorrect diagrams, showing apparently randomly scattered points, which they claimed showed cases in which $r_s = 1$ but $r \neq 1$. Some gave separate diagrams for r and r_s . Some candidates thought that the statement is true for a reason such as " r_s is r for ranks".	
			Total	7		



Question			Answer/Indicative content	Marks	Part marks and guidance	
26		i	$(a) S_{xx} = 3215 - \frac{179^2}{12}$ $(\text{= } 544.9167 \text{ or } \frac{6539}{12})$ $S_{yy} = 3565 - \frac{191^2}{12}$ $(\text{= } 524.9167 \text{ or } \frac{6299}{12})$ $S_{xy} = 3343 - \frac{179 \times 191}{12}$ $(\text{= } 493.9167 \text{ or } \frac{5927}{12})$	M1	Correct sub in a correct ① formula	
		i	$r = \frac{"493.9167"}{\sqrt{"544.9167" \times "524.9167"}}$	M1	Correct sub in 3 correct ① formulae and a correct r formula	Correct ans, (≥ 3 sf) no wking, M1M1A1
		i	= 0.924 (3 sf) or 0.9235.....	A1	Must see ≥ 3 sf	Ignore any comparison with 0.92 Examiner's Comments The vast majority of candidates answered this well, with the actual calculation being carried out accurately in most cases. A few truncated their answer to 3 significant figures, rather than rounding it, thereby losing a mark. There were very few cases of premature rounding of one or more ①-values.
		i	(b) Correlation does not imply causation or It depends on the time of year or Both depend on a third variable or There could be other factors	B1	Any answer which implies or is equiv to any of these	Allow without context. Ignore incorrect comments Examiner's Comments This is a simple, standard, and very common, question. Nevertheless, many candidates appeared not to be aware of the fact that correlation does not imply causation.



Question			Answer/Indicative content	Marks	Part marks and guidance	
	ii		(a) 'Increased' and Positive gradient or positive coeff of n or 'Output goes up by 0.6 each month' Both needed	B1	'Increased' and values of z shown as follows: at least 6 values or 1st and last values or 1st, or 2nd or 3rd or 4th and 9th or 10th or 11th or 12th ie 17.6 or 18.2 or 18.8 or 19.4 and 22.4 or 23 or 23.6 or 24.2	'Increased' and 'Value of 0.6n increases as n increases' NOT: 'Increased' and 'Value of z incr as n incr' 'z incr as no. of mths incr' Examiner's Comments Disappointingly few candidates gave the simple answer that the gradient of the regression line is positive and therefore the output increases. The majority actually calculated some of the values of the output. Some even calculated all 12 values! Many gave only the first two or three values, but this did not gain the mark because it does not comprise sufficient evidence that the output generally increased. In parts (ii)(b) and (ii)(c) many candidates used the unnecessarily long methods described below. They gained full marks if they obtained the correct answers, but the large amount of arithmetic involved meant that they were liable to make arithmetical errors.
	ii		(b) $\bar{n} = 6.5$ or $\frac{78}{12}$ oe seen	B1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
	ii		$\bar{z} = 0.6 \times '6.5' + 17$ alone, eg not $\div 12$ or $17 = \bar{z} - 0.6 \times "6.5"$ oe	M1	or $(0.6 \times 1 + 17 + 0.6 \times 2 + 17 + \dots 0.6 \times 12 + 17) \div 12$ oe or $'250.8' \div 12$ M1 ft their '6.5' only if comes from $\div 12$	Long method, all correct terms seen and $\div 12$ M1 NB ans 20.9 may not score the B1
	ii		$\bar{z} = 20.9$	A1	cao	Examiner's Comments Most candidates found $\bar{z}$ but, again disappointingly, many found $\bar{z}$ by using the equation of the regression line to find all twelve values of z rather than by substituting their value of $\bar{n}$.
	ii		(c) Total output = "20.9" $\times$ 12	M1	or $0.6 \times 1 + 17 + 0.6 \times 2 + 17 + \dots 0.6 \times 12 + 17$ oe or eg $\frac{88}{5} + \frac{91}{5} + \dots + \frac{121}{5}$ oe	Long method, all correct terms seen
	ii		251 (3 sf)	A1f	ft their (ii)(b)	Not ISW, eg 25100 scores A0, even if 251 seen Examiner's Comments Again, many candidates worked out all 12 values of the output and averaged them, rather than multiplying their answer to part (ii)(b) by 12.
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	-1	B1		Examiner's Comments Most candidates gave the correct answer, although a few gave answers such as "1" or "-0.9".
		ii	5 pts (or line or curve or zigzag) such that: grad always +ve (not vertical)	B1	Allow ≥ 4 pts	eg st line, +ve grad, B1B0 If crosses and curve or line, mark crosses
		ii	not in st line.	B1 dep	dep 1st B1 Must be clearly intended not to be st line	SC Some segments vertical (not all) B0B1 Examiner's Comments This question seemed to catch many candidates unawares and many failed to see the point about the difference between the two coefficients. Some gave five points in a straight line. Some gave apparently randomly placed points having a general tendency towards positive linear correlation, but with y not always increasing with x. A few appeared to be trying to give a correct answer, but inadvertently placed one point vertically above another, or horizontally on the same level as another.
		iii	$\begin{array}{cccccc} x & 1 & 2 & 3 & 4 & 5 \\ y & 1 & 2 & 3 & 5 & 4 \end{array}$ or $\begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 \\ 2 & 1 & 3 & 4 & 5 \end{array}$	M1	Attempt ranks	
		iii	Allow both sets reversed	A1	Correct ranks	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\Sigma d^2 = 2$	M1	dep 1st M1 Correct method for Σd^2 ; ft their ranks NB $\Sigma d^2 = 2^2 = 4$: M0	$\Sigma x = \Sigma y = 15$ $\Sigma x^2 = \Sigma y^2 = 55$ $\Sigma xy = 54$ $\textcircled{1} x = \textcircled{1} y = 55 - (15^2 \div 5) = 10$ $\textcircled{1}$ $xy = 54 - (15^2 \div 5) = 9$ correct method for one $\textcircled{1}$ M1 $r_s = \frac{9}{\sqrt{10 \times 10}}$ fully correct method M1 ans 0.9, no wking: full marks ans -0.9 <u>with wking</u> may get M1M1M1 ans -0.9, no wking: no marks
		iii	$r_s = 1 - \frac{6 \times 2^2}{5(5^2 - 1)}$	M1	Sub their Σd^2 into correct formula, dep M1M1, eg not using $\Sigma d^2 = 2^2 = 4$	= 0.9 A1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	= 0.9	A1		ans 0.9, no wking: full marks ans -0.9 with wking may get M1M1M1 ans -0.9, no wking: no marks Examiner's Comments Faced with no values of ranks, a large minority of candidates were unable to start this question, not realising that the ranks can be read off the diagram. A few candidates postulated that $r_s = 1$, then deduced that $\sum d^2 = 20$, and substituted this into the formula and, predictably, obtained the answer $r_s = 1$. Some candidates attempted to read off coordinates, which is quite acceptable so long as the consequent values are ranked. But some candidates just used their raw coordinates to find $\sum d^2$ without first assigning ranks. A few candidates read off the ranks incorrectly and some ranked the x-values and y-values in opposite directions. Some candidates gave a value for $\sum d^2$ without showing any ranks. Many candidates found either $\sum d = 2$ or $\sum d^2 = 2$, and then squared the "2" before substituting in the formula for r_s.
			Total	8		



Question			Answer/Indicative content	Marks	Part marks and guidance	
28		i	$S_{xx} = 200.39 - \frac{33.7^2}{6}$		or 11.108 or 11.1 or $\frac{1333}{120}$	
		i	$S_{yy} = 28314 - \frac{410^2}{6}$		or 297.333 or 297 or $\frac{892}{3}$	
		i	$S_{xy} = 2313.9 - \frac{33.7 \times 410}{6}$	M1	or 11.067 or 11.1 or $\frac{166}{15}$	Correct sub in a correct ① formula or correct value of one ① seen
		i	$r = \frac{"11.067"}{\sqrt{"11.108" \times "297.333"}}$ = 0.193 (3 sf)	M1		Correct sub in 3 correct ① formulae and a correct r formula
		i		A1	Examiner's Comments This question was well-answered, with only a few candidates showing some of the errors mentioned above, including giving only an answer correct to two significant figures, rather than the standard three.	No working: 0.193 M1M1A1 Ignore comment about $0 < r < 0.2$

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(For these 6 clubs) No / little / poor / weak oe Relationship / corr'n / link oe between (top) salaries and no. of points	B1	Allow without "For these 6 clubs" & "top" or "no strong corr'n between etc" In context. Examiner's Comments This was also well-answered, although a significant minority lost the mark by not setting their answer in context. Some others lost the mark by referring to goals instead of points. There were some imaginative answers such as 'The value of r shows whether the players were worth what they got paid' but these did not gain the mark. Some candidates failed to understand the significance of the small value of r, giving answers such as 'There is a small correlation between the top salary and points scored. As the top salary increases, the points scored increases.'	Allow "Salary has little effect on points" Ignore all else including "positive" NOT if use "goals" instead of "points"

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Extrapolation oe Corr'n poor / weak or no rel'nship / link oe or Points not close to line Small sample or only (top) 6 clubs oe	B1B1	Outside range of values. Salary is less than the others. r small or r close to 0 or r not close to 1 or Results do not correlate well Any two; allow without context Examiner's Comments Most candidates gave at least one of the two rote answers (Extrapolation and poor correlation), but many gave two answers that amounted to the same thing. Some gave the rote answer 'Correlation does not imply causation', which did not gain a mark.	NOT "Corr'n does not imply causation" NOT "Could be other factors" NOT if use "goals" instead of "points"
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
29		i	Year 80 81 82 83 84 85 86 87 88 Age 1 2 3 4 5 6 7 8 9 Quality 1 3 4 2 5 6 8 7 9		Y 80 83 81 82 84 85 87 86 88 Q 1 2 3 4 5 6 7 8 9 A 1 4 2 3 5 6 8 7 9	One set reversed, max 4 mks, eg Y 80 81 82 83 84 85 86 87 88 A 9 8 7 6 5 4 3 2 1 Q 1 3 4 2 5 6 8 7 9
		i	Attempt ranks	M1		or similar Attempt ranks M1
		i	Correct ranks	A1	Allow both sets of ranks reversed	Incorrect ranks A0
		i	Attempt $\Sigma d^2 (= 8)$	M1		Attempt $\Sigma d^2 (= 232)$ M1
		i	$1 - \frac{6 \times "8"}{9 \times (81-1)}$	M1		$1 - \frac{6 \times "232"}{9 \times (81-1)}$ M1
		i	$= \frac{14}{15}$ or 0.93 or 0.933 (3 sf)	A1	NB 0.93 is correct Examiner's Comments This question was well-answered by many. The mistakes mentioned above were seen infrequently. Occasionally a candidate made a mistake in the ranking. Some ignored the instruction that years should be ranked from the earliest (1) to the latest (9).	$-\frac{14}{15}$ or -0.93 or -0.933 (3 sf) A1

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Older is better oe or newer is worse oe As age increases, quality increases Must imply older is better oe, ie “good (or positive) corr’n between age and quality” is not enough	B1	No ft from (i) –0.933 in (i) leads to same conclusion as + 0.933 in (ii) Nothing contradictory seen, ie NOT ignore all else In context; no need to include “rank” Examiner’s Comments Many answers omitted the context. Attempts at rote-learned answers often failed to make it clear that the good rank correlation suggests that, generally speaking, ‘Older is better.’	NOT as year increases quality increases NOT High / strong / good corr’n / agreement / rel’nship between age and quality oe
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	$S_{xx} = 20400 - \frac{360^2}{8} \quad (= 4200)$			
			$S_{yy} = 6.88 - \frac{6.8^2}{8} \quad (= 1.1)$			
		i	$S_{xy} = 241 - \frac{360 \times 6.8}{8} \quad (= -65)$	M1	Correct sub in a correct ① formula	
		i	$r = \frac{"-65"}{\sqrt{"4200" \times "1.1"}}$	M1	Correct sub in 3 correct ① formulae and a correct r formula	
		i	$= -0.956 \text{ (3 sf)}$	A1	Correct ans with no working M2A1	Ignore comment about $-1 < r < -0.9$
					Examiner's Comments This part was well answered by most candidates. A few made the errors mentioned above. A significant number made an arithmetical slip leading to ①_{xx} = -66 instead of -65. Over rounding was frequently seen in this question, i.e. '0.96'.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	eg As you move further away, prices drop	B1	High prices go with short distances oe Allow "Strong (or high or good or equiv) neg corr'n between price and distance" Examiner's Comments Many candidates failed to mention the context. Others stated that the value of r shows "negative correlation" between price and distance, omitting to include "strong" or an equivalent. A few candidates simply discussed the nearness of points to a line of regression. Some candidates thought that a high negative value of r means that there is very little correlation.	Both variables must be in context; miles & £ enough Ignore all else, even if incorrect NOT just neg corr'n between price & dist
		iii	None	B1	Examiner's Comments Most candidates answered this part correctly, although some did not appreciate the difference between "affect" and "effect". They were not penalised for this non-mathematical error.	Ignore all else, even if incorrect
		iv	$b = \frac{-65}{4200}$ ($= -0.0154762$)	M1	ft their $\textcircled{1}_{xy}$ & $\textcircled{1}_{xx}$ from (i) for M-marks only	or fresh start correct method
		iv	$Y - \frac{6.8}{8} = -0.0154762(x - \frac{360}{8})$ oe	M1	or $a = \frac{6.8}{8} + "0.0154762" \times \frac{360}{8}$ oe	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y = -0.0155x + 1.55$ (3 sf) oe or $y = \frac{433}{280} - \frac{13}{840}x$ oe	A1	allow $y = -0.015x + 1.5$ (or figs which round to these) (NOT $y = -0.016x + 1.6$ NOT $y = -0.02x + 1.5$) Correct ans with no working M2A1 Examiner's Comments Like part (i) this was generally well answered. Some candidates started from scratch, not appreciating that they had already found the values of $\textcircled{1}_{xx}$ and $\textcircled{1}_{xy}$ in part (i). A few of these, having found the $\textcircled{1}$ values correctly in part (i), tried to use the formula for b from MF1 ($b =$ $\frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$) which they clearly did not understand. Some used incorrect values such as $\textcircled{1}_{xy} = 241$. Others made errors in the sign or the decimal point or rounding. Subtracting a negative to find 'a' caused problems for some candidates.	Must have "y = " Allow figures in equn which round to the correct figures to either 3 sf or 2 sf, Correct ans with no working M2A1 even if they result from arith errors.



Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	Values of x are chosen beforehand or x is independent or controlled	B1	x is fixed or given or set or predetermined oe Examiner's Comments Many candidates answered correctly. A few gave inadequate statements, such as that y is dependent on x. Others gave incorrect statements such as that y is independent or that x depends on y. Some wrote "because r is close to -1". A large number stated that this would be a better method because no rearrangement would be necessary.	Not "x is constant." Not just "y depends on x" Ignore all other, even if incorrect
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
31		i	6 5 4 3 2 1	B1	Examiner's Comments Almost all candidates gave the correct answer. A very few gave 1, 2, 3, 4, 5, 6. An even smaller number gave 2, 1, 4, 3, 6, 5 or 0, 1, 2, 3, 4, 5.	
		ii	$\Sigma d^2 = 0$ for first 6 terms	M1	May be implied by use of $\Sigma d^2 = 2$	using ranks from (i) can score 2nd M1 only
		ii	$\Sigma d^2 = 2$	B1		
		ii	$1 - \frac{6 \sum d^2}{8(8^2 - 1)}$	M1	ft their $\Sigma d^2 (\neq 0)$	
		ii	$= \frac{41}{42}$ or 0.976 (3 sf)	A1	Examiner's Comments A good number of candidates understood this rather non-standard question and gained full marks. However, the errors mentioned above were seen. But the most common error was to take the ranks from part (i) and use them in this part, giving $r_s = \frac{1}{7}$. Some considered only the two teams G and H. A few tried to work backwards from an r-value of -1 (or even 1) and find d^2 . Then they added 2 to this and proceeded to use the formula. $n = 6$ was used by a few candidates.	
			Total	5		